

Accelerating Pediatric Oncology Drug Development: Advances in Clinical Pharmacology, Trial Design, Non-Clinical Evidence, and Regulatory Science

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Abstract

Pediatric oncology drug development remains uniquely challenging due to the rarity and biological heterogeneity of childhood cancers, ethical considerations in trial conduct, and the limited feasibility of large, randomized studies. Despite these barriers, recent years have seen a notable acceleration in the approval of oncology therapies for pediatric populations, driven by advances in molecularly targeted treatments, evolving regulatory requirements, and innovation in trial design. Reducing nonclinical data requirements and increasing the adaptation of model-informed drug development approaches are also contributing to advances in pediatric oncology treatment by supporting dose selection, optimizing study design, and reducing unnecessary patient burden. In this review, recent regulatory requirements from the United States, the EU, and other key regions on pediatric oncology drug development are discussed. Thirty-three drugs with oncology indications in pediatric populations approved by the US FDA between 2018 and 2025 are reviewed. These approvals provide examples of how nonclinical and clinical data were generated with a focus on strategies for dose-finding and justification. Common challenges and considerations related to clinical operations and formulation development in pediatric populations and the emerging use of real-world data, external controls, and artificial intelligence/machine learning are also discussed.

Keywords

clinical pharmacology, clinical trial design, model-informed drug development, oncology, pediatric, real-world evidence

Introduction

The advancement in the understanding of cancer biology, novel treatment modalities, personalized medicine, and supportive care has increased the 5-year survival rate of pediatric cancers from 63% in the mid-1970s to 87% in 2015–2021 in the United States.^{1–3} As an example, for acute lymphocytic leukemia (ALL), the most common type of leukemia in children (ages 0–14 years) and adolescents (ages 15–19 years), the 5-year survival rate has steadily increased from 14% in the late 1960s to an impressive 90% in 2026.^{4,5} Despite these successes, cancer remains the leading disease-related cause of death in children and adolescents with an estimated 9680 and 5660 new cases and 1090 and 730 deaths, respectively in 2026,⁵ highlighting the need for continued research and innovative treatment approaches.

In recent years, there has been a notable increase in the number of approvals of oncology drugs for pediatric populations by the US Food and Drug Administration (FDA). FDA approvals have exhibited a sustained upward trajectory since 2017, consistently exceeding 5 approvals per year, with more than 15 distinct drug-indication approvals recorded in 2025 alone (Figure 1). The current approval landscape

is dominated by small molecules and monoclonal antibodies (mAbs), while newer modalities such as antibody drug conjugates (ADCs), bispecific antibodies, radiopharmaceuticals, and cell therapies remain relatively limited. Drug approvals span a broad range of pediatric malignancies, led by leukemia but extending to multiple solid and other rare cancers, indicating expanding therapeutic coverage across pediatric cancers. Importantly, most approved drugs include indications for infants, children, and adolescents, whereas

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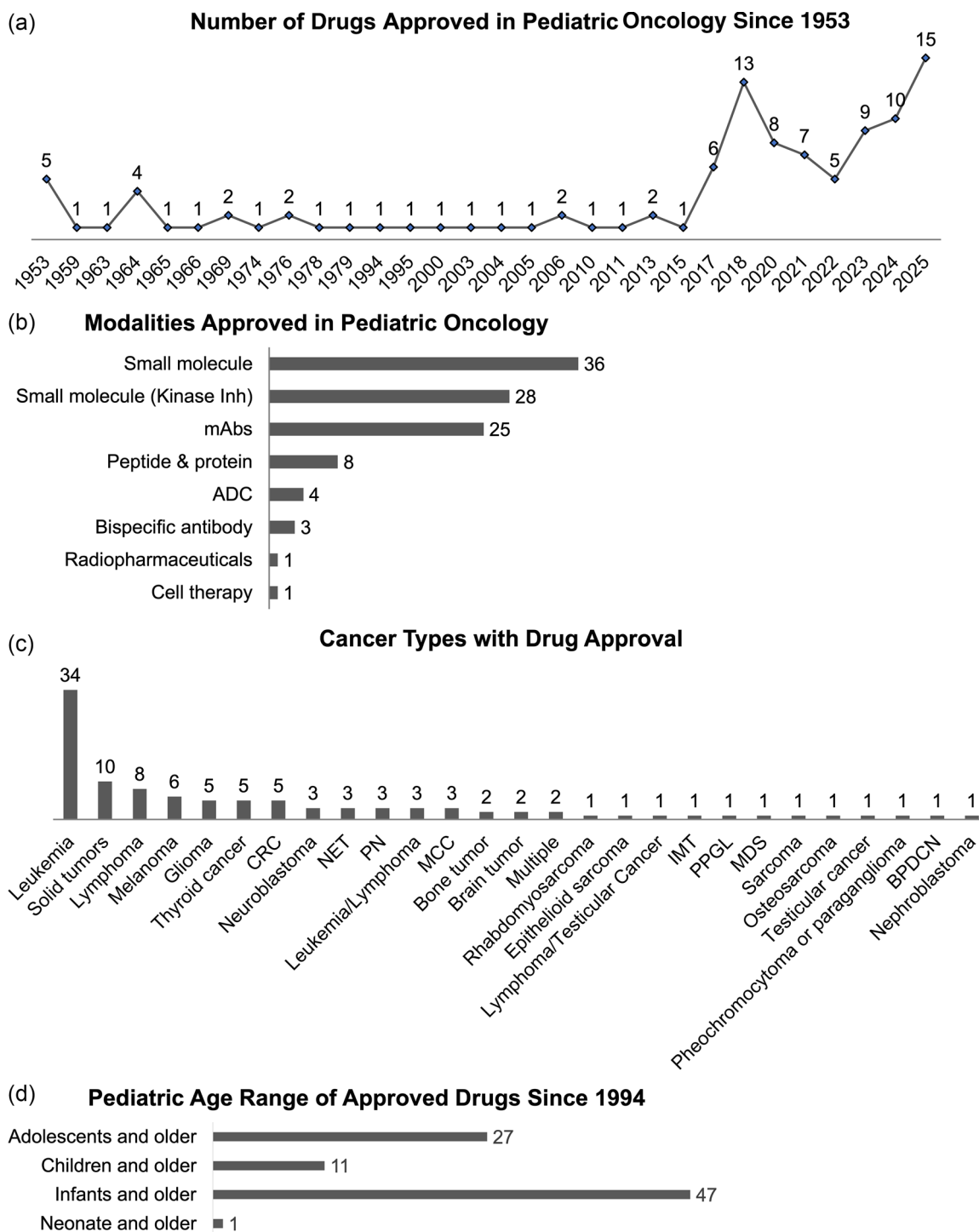


Figure 1. Trend in US FDA pediatric oncology drug approvals. (a) Annual number of pediatric drug approvals since 1953, showing a marked increase after 2017. (b) Distribution of approved modalities, dominated by small molecules, including kinase inhibitors and monoclonal antibodies. (c) Pediatric cancer types with at least one approved drug, led by leukemia and solid tumors such as sarcoma. (d) Approved drugs by pediatric age range, with most indications in infants and adolescents. Neonates: birth to 28 days; infants: 29 days to less than 2 years; children: 2 years to less than 12 years; adolescents: 12 to less than 17 years. ADC, antibody drug conjugate; BPDCN, blastic plasmacytoid dendritic cell neoplasm; CRC, colorectal cancer; Inh, inhibitor; mAbs, monoclonal antibodies; NETs, neuroendocrine tumors; IMT, inflammatory myofibroblastic tumor; MCC, Merkel cell carcinoma; MDS, myelodysplastic syndrome; PN, plexiform neurofibromas; PPGL, pheochromocytoma or paraganglioma.

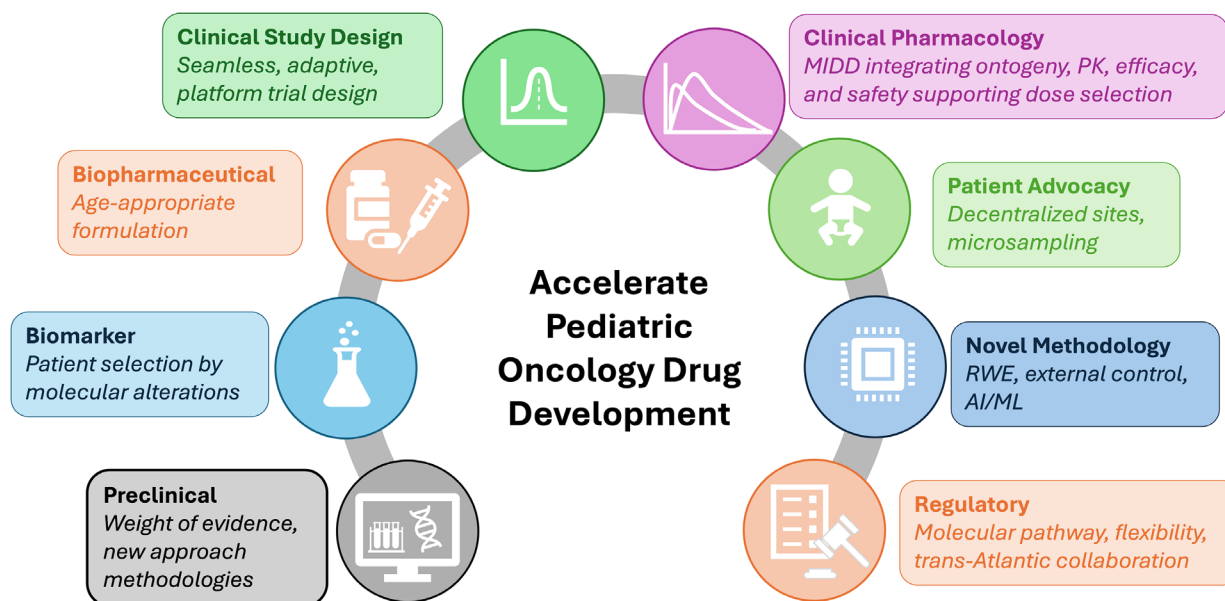


Figure 2. Recent advancements in accelerating pediatric oncology drug development. AI, artificial intelligence; MIDD, model-informed drug development; ML, machine learning; PK, pharmacokinetic; RWE, real-world evidence.

approvals that explicitly include neonates remain uncommon, underscoring persistent evidence gaps in the youngest and most developmentally vulnerable patient population.

This surge in drug approvals can largely be attributed to the strategic requirements and incentives established by legislation in the United States, such as the Research to Accelerate Cures and Equity (RACE) for Children Act in 2017,^{6,7} and regulatory requirements such as initial Pediatric Study Plans (iPSPs) by the FDA. These requirements have played a pivotal role in moving the initiation of pediatric study from a median of 0.04 years after adult approval before the RACE Act to a median of 2.8 years before the adult approval after the RACE Act,⁸ significantly fostering the development of pediatric oncology therapeutics.⁹ As a result, this article focuses on FDA approvals, while acknowledging the important role existing legislation outside of the US plays in support of global pediatric drug development.

The importance of clinical pharmacology in pediatric oncology was previously articulated by Shebley et al in 2019.¹⁰ Building upon this foundation, this review synthesizes key clinical pharmacology considerations derived from selected oncology drugs approved between 2018 and September 2025 (Table 1, see **Clinical Study Design** section for the selection criteria). In addition, it offers a detailed roadmap for stakeholders in pediatric oncology, encompassing essential aspects of regulatory engagement, preclinical data requirements, practical considerations for pediatric oncology trials, advances in quantitative pharmacology, and the emerging role of novel approaches and technologies in pediatric oncology

drug development (Figure 2). Together, these evolving pediatric regulatory requirements have elevated clinical pharmacology to play a central role in pediatric oncology drug development, with exposure matching, pharmacokinetic/pharmacodynamic (PK/PD) modeling, and other quantitative approaches increasingly forming the evidentiary basis for regulatory decision-making.

Regulatory Requirement

Key Regulatory Frameworks and Global Considerations

Pediatric requirements have become an increasingly early and integral component of oncology drug development. The United States and the EU now apply markedly different regulatory triggers—target-based in the United States and condition-based in the EU—requiring sponsors to plan pediatric strategies proactively and coordinate globally from the outset.

United States: PREA, RACE, and BPCA

In the United States, the Pediatric Research Equity Act (PREA) requires pediatric assessments for most new drugs and biologics unless waived or deferred. Sponsors must submit an iPSP no later than 60 days after the End-of-Phase 2 meeting and before beginning pivotal trials, with the FDA expecting the iPSP to be agreed prior to New Drug Application (NDA)/Biologics License Application (BLA) submission.^{47,48} PREA allows full or partial waivers—including cases where pediatric studies would be impossible, unethical, or unlikely to be beneficial—and deferrals when studies must occur after adult data become available.^{47,49} Deferral requests must include specific milestones and timelines.^{47,49} For example, the development of

Table 1. Summary of Indication, Dosing Method, and Dose Justification in Selected Oncology Drugs With US FDA Approval in Pediatrics (N = 33) Between 2018 and September 2025

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
Anti-CTLA-4 mAb	Ipilimumab (Yervoy)	• IL CRC (MSI-H or dMMR) with nivolumab (04/08/2025) • Melanoma as a single agent or with nivolumab (05/25/2021) • 2L CRC (MSI-H or dMMR) with nivolumab (07/10/2018)	Original approval in adults (3/25/2011), expanded to 12 years and older	BW-based (1 or 3 mg/kg Q3W or 1 mg/kg Q6W, depending on indications)	BW-based (1 or 3 mg/kg Q3W or 1 mg/kg Q6W, depending on indications)	Similar disease biology in pediatrics $\geq$ 12 y/o with adults and similar mechanism of action of checkpoint inhibitor across ages allows extrapolation of efficacy and safety; drug exposures in pediatrics $\geq$ 12 y/o are comparable to adults at the recommended dose. ¹¹
Anti-PD-1 mAb	Nivolumab (Opdivo)	• 1L CRC (MSI-H or dMMR) with ipilimumab (04/08/2025) • 2L CRC (MSI-H or dMMR) (04/08/2025) • Melanoma (adjuvant) (10/13/2023) • Melanoma as a single agent or with ipilimumab (02/15/2023) • 2L CRC (MSI-H or dMMR) as a single agent or with ipilimumab (07/10/2018)	Original approval in adults (12/22/2014), expanded to 12 years and older	Weight-tiered with BW-based BW <40 kg: BW-based (3 mg/kg Q2W or 6 mg/kg Q4W) BW $\geq$ 40 kg: Fixed (240 mg Q2W or 480 mg Q4W)		Similar disease biology in pediatrics $\geq$ 12 y/o with adults and similar mechanism of action of checkpoint inhibitor across ages allows extrapolation of efficacy and safety; drug exposure in pediatrics $\geq$ 12 y/o within the range in adults at the same dose. ¹²
Anti-PD-1 mAb and anti-LAG-3 mAb	Nivolumab and relatlimab-rmbw (Opduvalag)	Melanoma (03/18/2022)	Original approval in 12 years and older (and BW > 40 kg)	Fixed 480 mg nivolumab and 160 mg relatlimab Q4W		Similar disease biology in pediatrics $\geq$ 12 y/o with adults and similar mechanism of action of checkpoint inhibitor across ages allows extrapolation of efficacy and safety; drug exposures in pediatrics $\geq$ 12 y/o and BW > 40 kg are expected to be in the range of adults; BW was a significant covariate and only one patient with BW <40 kg (14 y/o, BW 39.7kg). ¹³
Anti-PD-1 mAb	Pembrolizumab (Keytruda)	• Melanoma (12/03/2021) • cHL (10/14/2020) • Solid tumors (TMB-H/MSI-H or dMMR; 06/16/2020) • MCC (12/19/2018) • PMBCL (06/13/2018)	Original approval in adults (9/4/2014), expanded to 12 years and older for melanoma (no specific age cutoff for other indications)	BW-based with cap (2 mg/kg, up to 200 mg Q3W)	Fixed (200 mg Q3W or 400 mg Q6W)	Similar disease biology in pediatrics $\geq$ 12 y/o with adults; drug exposure in pediatrics $\geq$ 12 y/o within the range in adults at the same dose; efficacy and safety data demonstrated in pediatrics $\geq$ 12 y/o. ¹⁴

(Continued)

Table 1. (Continued)

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
Anti-PD-1 mAb	Pembrolizumab and berahyaluronidase alfa-pmpH (Keytruda Qlex)	- Solid tumors (TMB-H/MSI-H or dMMR; 09/19/2025) - Melanoma (09/19/2025) - MCC (09/19/2025)	Original approval in 12 years and older (and BW > 40 kg)	Fixed 395 mg pembrolizumab and 4800 units berahyaluronidase alfa Q3W or	790 mg pembrolizumab and 9600 units berahyaluronidase alfa Q6W	PK in pediatrics ≥ 12 y/o is predicted to be within the range of adults using PopPK; comparable PK and efficacy between pembrolizumab SC and IV in adults; flat ER for pembrolizumab IV; ¹⁵ Efficacy extrapolated from adult study; safety and dosimetry data from adolescent data; PK matching between adolescents and adults. ¹⁶
Radiolabeled somatostatin analog	Lutetium Lu 177 dotatate (Lutathera)	GEP-NETs (04/23/2024)	Original approval in adults (1/26/2018), expanded to 12 years and older	Fixed (200 mCi Q8W, total of 4 doses)		Efficacy extrapolated from adult study; safety and dosimetry data from adolescent data; PK matching between adolescents and adults. ¹⁶
Radioactive diagnostic	Iobenguane I-131 (Azedra)	PPGL (07/30/2018)	Original approval in 12 years and older	Weight-tiered with BW-based Dosimetric: > 50 kg: 5-6 mCi; ≤ 50 kg: 0.1 mCi Therapeutic: > 62.5 kg: 500 mCi; ≤ 62.5 kg: 8 mCi/kg		Safety extrapolated from neuroblastoma study; efficacy extrapolated from adult study; PK matching between adolescents and adults when using BW-based dosing. ¹⁷
EPAS1 (HIF2A) inhibitor	Belzutifan (Welireg)	PPGL (05/14/2025)	Original approval in adults (8/13/2021), expanded to 12 years and older	Weight-tiered (QD) ≥ 40 kg: 120 mg < 40 kg: 80 mg	Fixed (120 mg QD)	Similar disease biology in adolescents and adults; PK matching between adolescents and adults. ¹⁸
Methyltransferase inhibitor	Tazemetostat (Tazverik)	Epithelioid sarcoma (01/23/2020)	Original approval in 16 years and older	Fixed (800 mg BID)		Similar disease biology in adolescents and adults; age is not a significant covariate on CL; safety exposure margins overlapped with adults; secondary T-cell lymphoblastic lymphoma increased at high exposures in juvenile animals. ¹⁹
IDH1, IDH2 inhibitor	Vorasidenib (Vorango)	Astrocytoma or oligodendroglioma with susceptible IDH-1 or IDH-2 mutation (08/06/2024)	Original approval in 12 years and older	Weight-tiered (≥ 40 kg: 40 mg QD; < 40 kg: 20 mg QD)		Similar disease biology in adolescents and adults; BW correlated with CL and used allometric scaling; PK matching adult dosing at 40 mg QD; efficacy and safety extrapolated from adults, not by a pediatric study. ²⁰
Kinase inhibitor	Cabozantinib (Cabometyx)	- epNET (03/26/2025) - pNET (03/26/2025) - DTC (09/17/2021)	Original approval in adults (8/26/2011), expanded to 12 years and older	Weight-tiered (≥ 40 kg: 60 mg QD; < 40 kg: 40 mg QD)		PK matching adult dosing at 60 mg QD; efficacy extrapolated from adult not by a pediatric efficacy study; No clear exposure-PFS trend; higher exposure correlated with Grade ≥ 3 AEs. ²¹

(Continued)

Table 1. (Continued)

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
Approved in Children and Older (N = 4)						
Anti-CD30 ADC	Brentuximab vedotin (Adcetris)	cHL (11/10/2022)	Original approval in adults (8/19/2011), expanded to 2 years and older	BW-based with cap (1.8 mg/kg up to 180 mg Q3W)	180 mg Q3W	Exposures in 12-17y/o were consistent with adults; no clinically significant effect (age, sex, race, baseline albumin) on PK after accounting for BW in 3-17 y/o; exposure-response supported the approved dose. ^{22,23}
Anti-PD-L1 mAb	Atezolizumab (Tecentriq)	ASPS (12/09/2022)	Original approval in adults (5/18/2016), expanded to 2 years and older	BW-based with cap (15 mg/kg up to 1200 mg Q3W)	Fixed (840 mg Q2W or 1200 mg Q3W or 1680 mg Q4W)	Similar disease biology in pediatric 2-11 y/o with adults and adolescents; similar mechanism of action of checkpoint inhibitor across ages allows extrapolation of efficacy and safety; PK matching between pediatric ≥ 2 y/o and adults. ^{24,25}
CD123 cytotoxin	Tagraxofusp-erzs (Elzonris)	BPDCN (12/21/2018)	Original approval in 2 years and older	BW-based (12 μ g/kg/day)		BW correlated with CL and V_d ; age is not a significant covariate after adjusting for BW for PK, and not a significant covariate for exposure-efficacy and safety, with a similar safety profile in pediatric and adults. ²⁶
Kinase inhibitor	Selpercatinib (Retevmo)	- Solid tumors (RET gene fusion) (05/29/2024) - Thyroid cancer (05/29/2024) - MTC (05/29/2024) - PN (04/10/2020)	Original approval in 12 years and older (05/08/2020), expanded to 2 years and older	Fixed and BSA-tiered by age Age ≥ 12 years: Fixed <50 kg: 120 mg BID ≥ 50 kg: 160 mg BID Age 2 to 12 years: BSA-tiered TID or BID (not recommended for BSA <0.33 m ²) (Tablets may be dispersed for patients unable to swallow; capsules.)		RET-driven cancers have similar disease progression across age groups; adolescents had similar PK and safety to adults; BW was a significant covariate in PopPK but not sex, race, and age; QT prolongation risk is higher for BW ≥ 50 kg. Exposure matching to adults (AUC and C_{max}) to support pediatric dosing. ^{27,28}
Approved in infants and older (N = 17)						
Anti-CD30 ADC	Gemtuzumab ozogamicin (Mylotarg)	Newly diagnosed or relapse or refractory AML (CD33+) (06/16/2020)	Original approval in adults (5/17/2000), expanded to 1 month and older (newly diagnosed) or 2 years and older (relapse or refractory)	BSA and BW-based by age Newly diagnosed (with chemo) BSA ≥ 0.6 m ² : 3 mg/m ² (once) BSA <0.6m ² : 0.1 mg/kg (once) Relapse or refractory 3 mg/m ² on D11/4/7	BW-based Newly diagnosed (with chemo) 3 mg/m ² on D11/4/7 Relapse or refractory 3 mg/m ² on D11/4/7	Age, race, sex, renal impairment (mild or moderate), or hepatic impairment (mild) had no impact on PK; safety and efficacy were no different by age; near-maximal CD33 target saturation occurs at doses ≥ 2 mg/m ² . ²⁹

(Continued)

Table 1. (Continued)

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
Anti-CD22 ADC	Inotuzumab ozogamicin (Besponsa)	ALL (CD22+ B-cell precursor; 03/06/2024)	Original approval in adults (8/17/2017), expanded to 1 year and older	BSA (Cycle 1: 0.8/0.5/0.5 mg/m ² on Day 1/8/15 Q3W; Subsequent cycles: Achieve CR or CRi: 0.5 mg/m ² on Day 1/8/15 Q3W or Not achieve CR or CRi: 0.8/0.5/0.5 mg/m ² on Day 1/8/15 Q3W)		Exposure increases with decreasing BW at the approved dose; about a 30% increase in AUC in pediatric > 1 y/o compared with adults; safety and efficacy demonstrated in Study W203581 (NCT02981628); higher incidence of liver test abnormalities in pediatric compared to adults. ³⁰
Bispecific Ab	Blinatumomab (Blincyto)	- ALL (CD19+Ph- B-cell precursor; 06/14/2024) - ALL (MRD≥0.1%; (03/29/2018)	Original approval in adults (12/03/2014), expanded to 1 month and older	Weight-tiered with BSA Induction (Cycle 1) BW <45kg: BSA-based (15 µg/m ² /day) BW ≥45kg: Fixed (28 µg/day) Consolidation (Cycles 2-4)		Age and gender had no clinical impact on PK; linear PK over 5-30 µg/m ² /day in pediatric; exposures were comparable in pediatric ≥ 1 month and adults at the BSA-adjusted dose; CRS mitigated by step-up dosing was consistent with adults. ³¹
Anti-GD2 mAb	Naxitamab-ggqk (Danyelza)	Neuroblastoma (with GM-CSF; 11/25/2020)	Original approval in 1 year and older	BW-based (3 mg/kg, up to 150 mg/day on Day 1/3/5 Q4W)		BW and baseline IFN-γ are significant covariates on CL (not age [1-34 y/o], sex, and race); exposures for BW > 50 kg are not clinically different from BW of 30-50 kg; safety and efficacy demonstrated in pediatrics at the approved dose, PMC to assess ADA status on PK, safety, and efficacy. ³²
Anti-CD20 mAb	Rituximab (Rituxan)	Non-Hodgkin's lymphoma (CD20+ DLBCL, BL, CLL, B-ALL; 12/02/2021)	Original approval in adults (11/26/1997), expanded to 6 months and older	BSA-based (375 mg/m ² , on Day 2 before and Day 1 of each of the two induction cycles and Day 1 of each of the two consolidation cycles, for a total of six doses)		Similar disease biology in pediatrics 6 months-1 y/o and pediatrics ≥ 1 y/o allows PK extrapolation; PK comparable between pediatric age 3-12 y/o and age 12-18 y/o; PK matching between pediatrics ≥ 3 y/o and adults. ³³
Asparagine enzyme	Calaspargase pegol—mknl (Asparlas)	ALL (12/20/2018)	Original approval in 1 month and older	BSA-based (2500 units/m ² Q3W)		BW correlated with CL and Vd; no clinically meaningful differences in PK across age groups when dosed by BSA; approved dose maintains nadir serum asparaginase activity. ³⁴

(Continued)

Table 1. (Continued)

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
Nucleoside Inhibitor	Azacitidine (Vidaza)	JMML (05/20/2022)	Original approval in adults (5/19/2004), expanded to 1 month and older	Age-based and/or BW-based Age 1 month to <1 year OR <10 kg: 2.5 mg/kg/day × 7 days Age ≥1 year AND ≥10 kg: 75 mg/m ² /day × 7 days Cycles every 28 days; 3-6 cycles maximum (IV, not substitute for oral)		PK matching between pediatrics and adults when dosing by BSA or BW-based dosing. ³⁵
Topo inhibitor and nucleoside inhibitor	Daunorubicin and cytarabine liposome (Vyxeos)	t-AML or AML-MRC (03/30/2021)	Original approval in adults (8/3/2017), expanded to 1 year and older	BSA Induction: daunorubicin 44 mg/m ² + cytarabine 100 mg/m ² Consolidation: daunorubicin 29 mg/m ² + cytarabine 65 mg/m ²		Total daunorubicin and cytarabine exposures in pediatrics were within the adult exposure range when dosed by BSA; comparable safety profile to adults; extrapolate efficacy from adults. ³⁶
CLPP; DRD2	Dordaviprone (Modeyso)	Glioma (H3 K27M mutation; 08/06/2025)	Original approval in 1 year and older	Weight-tiered (QW) 10 to <12.5 kg: 125 mg 12.5 to <27.5 kg: 250 mg 27.5 to <42.5 kg: 375 mg 42.5 to <52.5 kg: 500 mg ≥52.5 kg: 625 mg BSA (BID) >1.5: 768 mg 0.75-1.5: 576 mg 0.5-0.75: 384 mg 0.25-0.5: 192 mg (Tablets can be swallowed, chewed, or crushed and mixed with small liquid volumes. No food effect allows an unpredictable feeding schedule.)	Fixed (625 mg QW)	BW correlated with CL; pediatrics <1 y/o were excluded because CYP3A maturation is incomplete, making PK predictions unreliable; exposures (AUC and C _{max}) were similar to adults with weight-tiered dosing. ³⁷ No DLTs and MTD identified in Ph I study; efficacy appeared not dose-dependent; PK exposures within a twofold window, appropriate for a cytostatic drug across ages 1-19 years, sexes, races, and BSA ranges; long-term toxicology showed known risks manageable at pediatric dose (e.g., hearing changes and mild cytopenias) ³⁸
Ornithine Decarboxylase Inhibitor	Eflornithine (Iwifin)	HRNB (12/13/2023)	Original approval in 1 year and older	Weight-tiered with fixed and BSA (≥40 kg: 270 mg BID; <40 kg: 160 mg/m ² BID; dose reduced with strong CYP3A4 inhibitor)		BW correlated with CL; children <40 kg had ~30% higher PK; age did not independently impact PK after adjusting for BW; PBPK supported pediatric exposures (1-17 y/o) matching adults; flat exposure-efficacy but higher exposure correlated with increased safety risks; efficacy confirmed in the pivotal study. ³⁹
MEN1 inhibitor	Revumenib (Revufori)	Acute leukemia (KMT2A translocation; 11/15/2024)	Original approval in 1 year and older			

(Continued)

Table 1. (Continued)

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
DNA alkylating	Treosulfan (Grafapex)	- MDS (01/21/2025) - AML (01/21/2025)	Original approval in 1 year and older	BSA (10 g/m ² /day)		Similar disease biology and comparable MOA across age groups allow efficacy extrapolation from adults to 1 y/o and older using PopPK; pediatric safety was supported by clinical data, and PK increased modestly ($\leq 11\%$) in children with smaller BSA relative to adults. ⁴⁰ PK matching adult dosing with acceptable safety margin; BE established (similar AUC _{0-12h} and C _{max}) between capsules and granules to extend approval from 2 to 1 y/o. ⁴¹
MEK1 and MEK2 inhibitor	Selumetinib (Koselugo)	PN (09/10/2025)	Original approval in 2 years and older (4/10/2020), expanded to 1 year and older	BSA (25 mg/m ² BID) (capsules and oral granules; capsules and oral granules not recommended for BSA < 0.55 and 0.4 m ² , respectively)		BSA significant impact on pediatric PK; BSA dosing to correct exposure; maximum efficacy, and tolerability at approved dose. ⁴² BW correlated with CL; exposures were similar to adults with weight-tiered dosing. ⁴³
Kinase inhibitor	Crizotinib (Xalkori)	- IMT (ALK+; 07/14/2022) - ALCL (ALK+; 01/14/2021)	Original approval in adults (8/26/2011), expanded to 1 year and older	BSA (280 mg/m ² BID; capsules and oral pellets)	Fixed (250 mg BID)	
Kinase inhibitor	Dabrafenib (Tafinlar; as a single agent or in combination with trametinib)	- Solid tumors (BRAF V600E mutation; 08/31/2023) - LGG (03/16/2023)	Original approval in adults (5/29/2013), expanded to 6 years and older (6/22/2022), then to 1 year and older	Weight-tiered 26-37 kg: 75 mg 38-50 kg: 100 mg BID ≥ 51 kg: 150 mg BID (capsules or tablets for oral suspension)	Fixed (150 mg BID)	
Kinase inhibitor	Larotrectinib (Vitrakvi)	Solid tumors (NTRK gene fusion; 11/26/2018)	Original approval in adults and 1 month and older	BSA-tiered BSA ≥ 1.0 m ² : 100 mg BID BSA < 1.0 m ² : 100 mg/m ² BID (max 100 mg; capsules and oral solution)	Fixed (100 mg BID)	BSA dosing in pediatric achieved similar AUC across age strata (1 month to 18 years) and 90% IC ₅₀ ; flat exposure–efficacy and safety; no MTD was reached in the pediatric dose escalation study. ⁴⁴
Kinase inhibitor	Nilotinib (Tasigna)	CML (Ph+) (03/22/2018)	Original approval in adults (10/29/2007), expanded to 1 year and older	BSA 230 mg/m ² BID (max 400 mg) (Capsule contents may be dispersed in 1 tsp. applesauce; bioequivalent to intact capsule)	Fixed (newly diagnosed: 300 mg BID; resistant/intolerant: 400 mg BID)	BSA correlated with CL and was the primary driver for PK differences between adult and pediatric; steady-state exposure was within twofold of adult exposure at the approved dose; C _{min} comparable across pediatric age groups, settings (newly diagnosed vs resistant/intolerant), and studies. ⁴⁵

(Continued)

Table 1. (Continued)

Pharmacologic Class	Generic Name (Brand Name)	Oncology Indications in Pediatric Patients (Approval Date)	Approval History and Approved Pediatric Age Range	Pediatric Dosing (Formulation Consideration)	Adult Dosing (For the Same Indication)	Pediatric Dose Justification
Approved in Neonates and Older (N = 1)						
IFNG inhibitor mAb	Enapalumab-lzsg (Gamifant)	Primary HLH and HLH activation syndrome (11/20/2018)	Original approval in newborns and older	BW-based Primary HLH: 1 mg/kg twice per week HLH activation syndrome: Day 1: 6 mg/kg; Days 4 to 16: 3 mg/kg Q3D for 5 doses; Day 19 and onward: 3 mg/kg twice per week		BW but not others (age, sex, race, renal impairment, including dialysis, and hepatic impairment) was a significant covariate on PK; exposure–safety supported the dose; exposure–efficacy was inconclusive due to the dose-escalation scheme and high correlation between PK and PD; ⁴⁶

ADC, antibody drug conjugate; ALCL, anaplastic large cell lymphoma; ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; t-AML, therapy-related AML; AML-MRC, AML with myelodysplasia-related changes; ASPS, alveolar soft part sarcoma; B-AL, mature B-cell acute leukemia; BID, twice daily; BL, Burkitt lymphoma; BPDCL, blastic plasmacytoid dendritic cell neoplasm; BSA, body surface area; BW, body weight; cHL, classical Hodgkin lymphoma; CML, chronic myeloid leukemia; CRC, colorectal cancer; CTLA-4, human cytotoxic T-lymphocyte antigen 4; DLBCL, diffuse large B-cell lymphoma; dMMR, mismatch repair deficient; DTC, differentiated thyroid cancer; epNET, extra-pancreatic neuroendocrine tumors; GEP-NETs, gastroenteropancreatic neuroendocrine tumors; HLH, hemophagocytic lymphohistiocytosis; HRNB, high-risk neuroblastoma; IDH, isocitrate dehydrogenase; IMT, inflammatory myofibroblastic tumor; JMML, juvenile myelomonocytic leukemia; LAG-3, lymphocyte activation gene-3; LGG, low-grade glioma; mAb, monoclonal antibody; MCC, Merkel cell carcinoma; MDS, myelodysplastic syndrome; MRD, minimal residual disease; MSI-H, microsatellite instability high; MTC, medullary thyroid cancer; NTRK, neurotrophic tyrosine receptor kinase; PD-(L)1, programmed death receptor (ligand)-1; PMBCL, primary mediastinal large B-cell lymphoma; Ph, Philadelphia chromosome; PN, plexiform neurofibromas; pNET, pancreatic neuroendocrine tumors; PPGL, pheochromocytoma or paraganglioma; TMB-H, tumor mutational burden high.

nivolumab (a PD-1 inhibitor) with relatlimab-rmbw (a LAG-3 inhibitor) as a fixed-dose combination (FDC) received a partial waiver from the requirements of PREA for patients <12 years of age, presumably due to the rarity of melanoma in pediatrics, the indication the sponsor seeks approval for, despite the clinical relevance of LAG-3 expression across pediatric tumors.¹³

The RACE for Children Act was passed by the US Congress in 2017 and took effect on August 18, 2020, significantly expanding PREA requirements for oncology. RACE links pediatric obligations to a drug’s molecular target or mechanism of action if that target is relevant to pediatric cancer biology, irrespective of the adult indication.^{50,51} It also eliminated the orphan exemption for targeted oncology therapies, meaning many oncology programs now face pediatric obligations much earlier than historically anticipated.⁵⁰ PREA and RACE impose mandatory requirements but provide no exclusivity incentives.⁴⁸

In contrast, the Best Pharmaceuticals for Children Act (BPCA) provides a voluntary pathway in which sponsors may obtain six months of pediatric exclusivity by conducting FDA-requested studies. Historically, the FDA sometimes issued BPCA Written Requests (WRs) for studies that were also required under PREA, allowing sponsors to receive exclusivity for fulfilling mandatory pediatric obligations. However, current FDA guidance clarifies that the Agency no longer expects to issue WRs solely for studies or planned studies required under PREA. In oncology, this shift is reinforced by the RACE for Children Act, under which molecularly targeted pediatric studies are now mandated under PREA and constitute the primary driver of pediatric oncology drug development.^{49,52}

In addition to de novo development under PREA, RACE, and BPCA, lifecycle strategies such as the 505(b)(2) and 505(j) pathways—which enable the development of modified versions of approved drugs and generic medicines, respectively—can support the adaptation or repurposing of existing oncology therapies, including new formulations or dosing approaches, to expand availability of treatments in pediatric populations.

European Union: Paediatric Regulation and the PIP Framework

The EU Paediatric Regulation requires agreement on a pediatric investigation plan (PIP) with EMA’s Paediatric Committee (PDco) no later than completion of adult PK studies.⁵³ The PIP establishes binding pediatric measures, study designs, and timelines, including requested waivers and deferrals. Compliance is verified at marketing authorization and when sponsors seek new indications or line extensions.^{53,54}

Table 2. Key Differences Between the US and EU Regulatory Framework in Pediatric Drug Development

Topic	PREA (US)	RACE (US)	BPCA (US)	EU PIP
Type of requirement	Mandatory pediatric assessments	Mandatory (expands PREA)	Voluntary incentive program	Mandatory pediatric development plan
Primary trigger	Adult indication	Target/MOA relevant in pediatric cancers	FDA Written Request	Pediatric relevance of the condition
Timing	iPSP due 60 days after EOP2	Follows PREA timelines	No fixed timing	PIP agreed before adult PK completion
Waivers	Feasibility, ethics, lack of benefit	Same as PREA	Not applicable	Disease absent in children or no benefit
Deferrals	Permitted with milestones	Same as PREA	Not applicable	Permitted with strict deadlines
Incentives	None	None	Six-month pediatric exclusivity	SPC extension or 2-year orphan exclusivity

BPCA, the Best Pharmaceuticals for Children Act; EOP2, End-of-Phase 2; iPSP, initial Pediatric Study Plan; MOA, mechanism of action; PIP, Pediatric Investigation Plan; PK, pharmacokinetic; PREA, the Pediatric Research Equity Act; RACE, Research to Accelerate Cures and Equity; SPC, Supplementary Protection Certificate.

PDCO may grant product-specific or class waivers where the relevant disease does not occur in children or where evidence indicates the product would not be safe, effective, or provide meaningful benefit in children.⁵⁵ Deferrals allow pediatric studies to be initiated or completed after submission of the Marketing Authorization Application, but require clear deadlines and formal modification for any subsequent changes.^{53,54}

Unlike the US target-based RACE framework, EU pediatric requirements are condition-based. PDCO uses a structured hierarchy to define the “condition”, which may be broader or narrower than the proposed adult indication, depending on pediatric medical need.^{53,55} Completion of all agreed PIP measures can yield substantial incentives: a 6-month Supplementary Protection Certificate (SPC) extension for non-orphan medicines and two additional years of orphan market exclusivity for orphan products.⁵³

EMA has also formalized the stepwise PIP (sPIP) approach for exceptional cases involving significant scientific uncertainty. An sPIP allows agreement on minimum core components of the plan while deferring detailed pediatric elements to predefined milestones.⁵⁶ Lastly, the EU framework includes the Paediatric Use Marketing Authorisation (PUMA), which provides 10 years of data and market protection for off-patent medicines developed exclusively for pediatric use, thereby supporting the repurposing of established therapies to expand availability for pediatric populations.

Global Challenges and Strategies for Harmonization

Differences between US and EU frameworks (Table 2) create challenges for designing coherent global pediatric oncology strategies. US obligations are triggered by biological target relevance, whereas EU obligations depend on potential pediatric use within a defined condition.^{50,53,55} These discrepancies are particularly complex for rare molecular alterations, biomarker-defined populations, and combination regimens.

Given small and internationally dispersed pediatric oncology populations, early and coordinated regulatory engagement is essential. In the United States, sponsors may seek advice through established FDA meeting frameworks and oncology-specific interactions via the Oncology Center of Excellence.^{57,58} In the EU, while the PIP procedure is the definitive mechanism for determining pediatric requirements, EMA Scientific Advice can be used to discuss methodological issues such as extrapolation, formulation development, and study design.^{59,60} Such advice complements but does not replace PDCO decision-making.^{53,54,60}

TransAtlantic collaboration mechanisms play an important role. The FDA–EMA pediatric cluster is a regulator-only forum where agencies exchange views on topics such as extrapolation, target relevance, and justifications for waivers and deferrals; sponsors can be informed of the outcome of these cluster meetings via the common commentary process.⁶¹

Parallel Scientific Advice (PSA) enables sponsors to obtain coordinated input from the FDA and EMA on specific scientific questions and is particularly valuable for pediatric oncology programs involving complex endpoints, dose optimization, extrapolation strategies, or sequencing of pediatric versus adult studies.⁶²

Outside the United States and the EU, most jurisdictions do not impose formal pediatric requirements. For example, in Japan, recent amendments to the Pharmaceutical and Medical Devices Act and Pharmaceuticals and Medical Devices Agency (PMDA) notifications require sponsors developing adult oncology drugs to explicitly consider and document pediatric development plans during clinical trial consultations, reflecting an emerging expectation for early lifecycle pediatric planning despite the absence of binding approval conditions. Chinese health authorities encourage early consideration of pediatric development and have introduced several incentive-based policies to support this area.⁶³ Great Britain operates a national GB PIP

procedure similar to the EU system, while other regions generally rely on scientific advice and case-by-case determinations.⁶⁴

Nonclinical Evidence Generation

Animal models continue to play an important but increasingly contextualized role in preclinical pediatric oncology drug development, supporting investigation of tumor biology, evaluation of antitumor activity, and characterization of safety prior to clinical testing. Their use is expected to be scientifically justified and aligned with pediatric-specific disease biology, and incorporated within a broader, fit-for-purpose evidence strategy rather than relying on adult cancer paradigms. In this context, regulatory agencies emphasize that adequate proof-of-concept data—derived from integrated nonclinical and, where appropriate, clinical evidence—is required to justify initiation of pediatric oncology trials.⁶⁵

The FDA's pediatric oncology framework encourages early integration of pediatric considerations into development programs and prioritizes generation of mechanistically relevant and clinically translatable nonclinical data.^{51,66,67} Recent efforts by the FDA's Center for Drug Evaluation and Research (CDER) to modernize nonclinical testing—particularly through adoption of new approach methodologies (NAMs) and the 3Rs principles—replacement, reduction, and refinement—have further refined expectations for evidence generation.⁶⁸ Nonclinical packages are thus increasingly context-of-use driven, incorporating both conventional models and NAMs within a weight-of-evidence framework to support regulatory decision-making. In many cases, pediatric clinical studies may be supported by adult clinical data combined with targeted nonclinical studies in relevant models, particularly for therapies for advanced cancers, life-threatening hematologic disorders,⁶⁹ or well-characterized mechanisms.^{70,71}

Within this paradigm, juvenile animal studies (JAS) are no longer considered routine. Instead, their use is hypothesis-driven and limited to scenarios where a drug may affect developmental processes. When conducted, JAS should address specific mechanistic questions related to postnatal development and inform pediatric dose selection and long-term safety monitoring.⁶⁷ A systematic analysis of 74 FDA-approved pediatric products found that JAS findings resulted in direct changes to product labeling in only a small proportion of cases (8%), supporting a more targeted, question-driven application rather than routine use.⁷²

Although no specific model is mandated, sponsors are expected to justify the biological relevance and translational value of chosen systems, align endpoints with pediatric disease mechanisms, and incor-

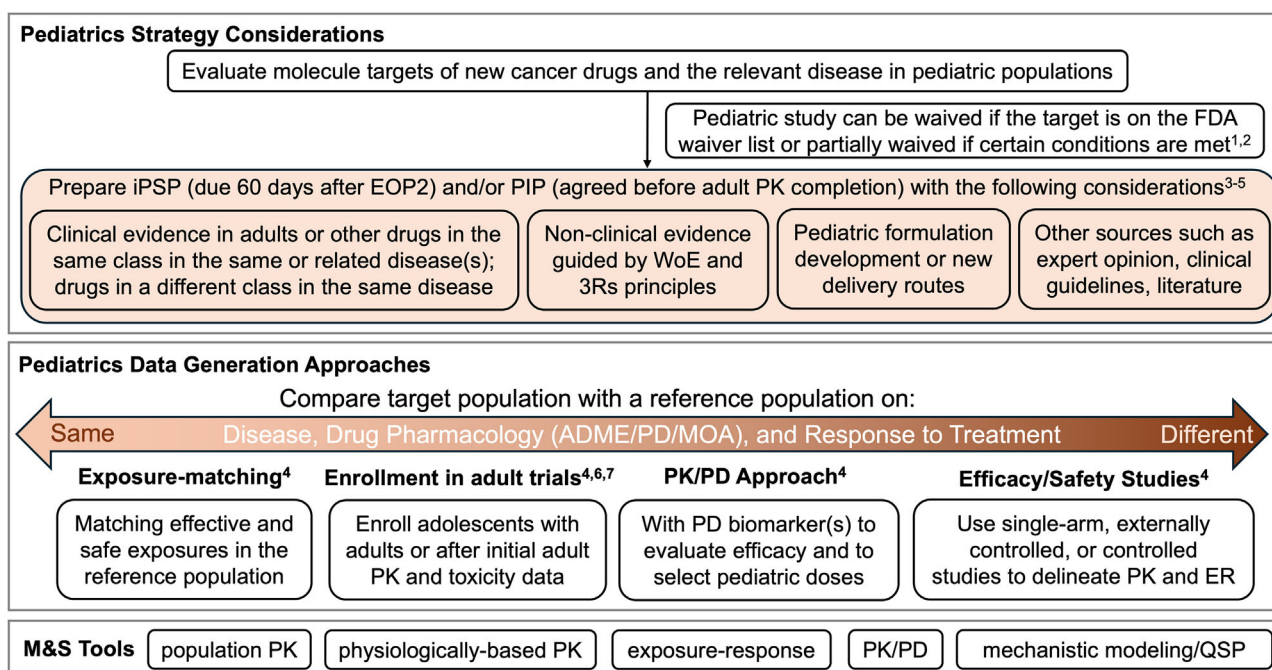
porate model-informed drug development (MIDD) approaches where appropriate. Increasingly, this includes integration of NAMs—such as human-relevant in vitro systems, in silico modeling, and functional assays—to improve prediction of pharmacology and safety. Initiatives such as the Pediatric High-Risk Cancer Preclinical Model Resource further support access to clinically relevant models and translational research.⁷³ Collectively, these strategies enable nonclinical data to be evaluated based on their capacity to inform pediatric dose selection, characterize risk, and support extrapolation under PREA and related frameworks.

NAMs are playing an increasingly central role by providing human-relevant data that complement, and in some cases reduce reliance on, traditional animal studies.⁷⁴ Consistent with the FDA's roadmap to modernize nonclinical testing, NAMs are being applied across pharmacology and safety assessments through integrated, fit-for-purpose approaches. Technologies such as organoids, organ-on-chip platforms, and computational modeling allow mechanistic interrogation of drug effects while improving translational relevance.^{75,76} In pediatric oncology, where limited patient populations, ethical constraints, and variable predictive value of animal models present challenges, these approaches offer advantages by enabling high-content testing with minimal patient-derived material. Together, the integration of NAMs with conventional models supports a more predictive, efficient, and ethically aligned nonclinical framework for pediatric oncology drug development.

Clinical Evidence Generation

Beyond the regulatory expectations for pediatric drug development outlined in the **Regulatory Requirement** section, the FDA and the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) provide guidance on practical approaches for generating clinical evidence to support pediatric drug approval (Figure 3). In particular, ICH E11A Pediatric Extrapolation clarifies the scientific principles and framework for extrapolating data from a reference population, commonly adults, to a pediatric target population.⁷⁷ Central to this framework is a systematic evaluation of similarities and differences in disease biology, drug pharmacology, and expected response to treatment.

Within this extrapolation framework, the regulator recognizes a continuum of development approaches tailored to the level of residual uncertainty. Exposure matching is appropriate when the disease course and mechanism of action are sufficiently similar between pediatric and adult populations, and when clinical benefit is primarily driven by systemic exposure such



¹US FDA. FDARA Implementation Guidance for Pediatric Studies of Molecularly Targeted Oncology Drugs: Amendments to Sec. 505B of the FD&C Act Guidance for Industry. May 2021

²EMA. Policy on the determination of the condition(s) for a Paediatric Investigation Plan/Waiver. January 2024

³US FDA. General Clinical Pharmacology Considerations for Pediatric Studies of Drugs, Including Biological Products Guidance for Industry. September 2022

⁴US FDA. E11A Pediatric Extrapolation Guidance for Industry. December 2024

⁵US FDA. S11 Nonclinical Safety Testing In Support of Development of Pediatric Pharmaceuticals Guidance for Industry. May 2021

⁶US FDA. Considerations for the Inclusion of Adolescent Patients in Adult Oncology Clinical Trials Guidance for Industry. March 2019

⁷For selected cancer types, for example, soft tissue and bone sarcoma, CNS tumor, leukemia/lymphoma, and melanoma

Figure 3. Integrated evidence generation strategy leading to pediatric oncology drug approval. ADME, absorption, distribution, metabolism, and excretion; EOP2, End-of-Phase 2; ER, exposure–response; iPSP, initial Pediatric Study Plan; M&S, modeling and simulation; MIDD, model-informed drug development; MOA, mechanism of action; PD, pharmacodynamic; PIP, Pediatric Investigation Plan; PK, pharmacokinetic; QSP, quantitative system pharmacology; 3Rs, replacement, reduction, and refinement; WoE, weight of evidence.

that achieving adult–like exposure in pediatric patients is expected to confer comparable efficacy. Enrollment of adolescents in adult oncology trials is also encouraged when cancers that are biologically similar to adult malignancies or share the same molecular target and body size–adjusted PK is expected to be similar to adults, enabling earlier access and generation of PK and safety data within adult programs. For younger pediatric populations, where maturational changes in absorption, distribution, metabolism, and excretion (ADME) may meaningfully impact exposure, population PK (PopPK) and physiologically based PK (PBPK) modeling are particularly appropriate to inform initial dose selection, account for developmental physiology, and support dose refinement prior to or in conjunction with limited pediatric PK data collection. (See **Clinical Study Design** and **Model-Informed Dosing Strategies** sections for examples of each approach.)

When exposure matching alone does not adequately support extrapolation, PK/PD–based approaches using biologically relevant biomarkers can confirm pharmacologic activity and guide pediatric dose selection. In contrast, when disease course, response, or endpoint relevance differs between adults and children, pediatric

efficacy and safety studies are needed, often using single–arm or externally controlled designs because randomized controlled trials may not be feasible due to rarity or ethical concerns. For ultra–rare pediatric malignancies, the FDA may also accept external controls or real–world data (RWD) to contextualize treatment effects when randomized trials are not feasible. (See **Novel Approaches and Data Type** section for further information.)

Ethical and Operational Considerations of Clinical Studies

Pediatric oncology research poses unique ethical challenges because the affected population is small, vulnerable, and faces life-threatening illnesses with limited alternatives. Clinical pharmacology decisions—dose selection, sampling intensity, extrapolation, and study design—are inseparable from these ethical considerations. Because most pediatric tumors are rare and geographically dispersed, traditional development pathways are often infeasible. Seamless or platform designs can reduce operational burden by consolidating infrastructure, sharing control arms, and enabling

response-adaptive decisions (see **Clinical Study Design** section).⁷⁸

Avoiding unnecessary pediatric trials is an important ethical topic. The 2022 Pediatric Oncology Subcommittee of the Oncologic Drugs Advisory Committee (ODAC) underscored the limited value and potential harm of duplicative trials with “next-in-class” agents when class pharmacology, exposure–response, and safety are already well characterized, and no plausible pediatric differentiation is expected. In such cases, selective waivers or streamlined evaluations may be warranted. Conversely, sponsors should demonstrate clear biological or clinical differentiation to justify additional pediatric exposure. These positions argue for prioritizing pediatric studies with the highest probability of benefit and the lowest risk of redundancy. These ethical considerations closely mirror the scientific and ethical criteria applied by regulators when evaluating requests for pediatric waivers, deferrals, or streamlined development programs under PREA and the EU Pediatric Regulation.

Dose-finding and combinations also raise specific ethical tensions. Beginning with monotherapy may characterize first-in-kid PK/PD and toxicity attribution but can expose kids to suboptimal antitumor activity in life-threatening settings. Starting directly with standard of care (SOC) combinations better reflects real-world practice but complicates causality for adverse events and intensifies drug–drug interaction risks. MIDD—PopPK, PBPK, and exposure–response modeling—can ethically de-risk both strategies by supporting biologically plausible starting doses, anticipating drug–drug interactions (DDIs), and shortening periods of non-therapeutic exposure.⁷⁹

Limited blood volumes in infants and young children pose procedural burden, mandating consideration of sparse sampling, microsampling (e.g., capillary or dried blood spot through devices like the Tasso OnDemand kit for pediatric or Neoteryx Mitra blood collection kit), and high-sensitivity assays. Under such constraints, model-based approaches generally outperform noncompartmental analysis for PK parameter precision, enabling optimal sampling windows through D-optimality approaches,^{80,81} smaller cohorts, and principled borrowing across age strata.

Prudent extrapolation is central to ethical study design. When pediatric tumors diverge biologically from adult disease, adult data should inform rather than dictate dose or benefit–risk assessments. On the other hand, for molecularly defined, tissue-agnostic targets with comparable biology across ages, exposure matching can be sufficient to select pediatric doses and support labeling.

Finally, the timing of pediatric initiation must weigh early access against uncertainty in developmental

safety. Juvenile animal findings can inform monitoring plans, but their translational precision is limited. Regulators encourage integrating nonclinical signals with human exposure margins, PBPK modeling considering ontogeny, and long-term follow-up, rather than relying on animal data alone.⁷⁹

Clinical Study Design

Review of Oncology Drugs with Approval in Pediatric Populations

To reflect recent clinical study considerations, this review focused on oncology drugs approved by the FDA in pediatric populations between January 2018 and September 2025. During this period, a total of 43 unique oncology drugs were approved, regardless of the number of indications each drug was approved for. For drugs within the same pharmacologic class and with the same pediatric indications, the drug with a wider age approval range or the drug with an earlier approval date was included. As such, 33 drugs were included (Table 1), and 10 drugs (asparaginase *Erwinia chrysanthemi* recombinant, mirdametinib, bosutinib, dasatinib, entrectinib, pralsetinib, repotrectinib, selumetinib, tovorafenib, and trametinib) were excluded in this review.

Understanding why oncology drugs approved in adult populations fail to demonstrate efficacy and/or safety in pediatric populations can be instructive. A recent review on drugs approved by the FDA between 2015 and 2022 in non-oncology therapeutic areas in adult populations identified that 15% (32 in 211) of drugs did not receive approval in pediatric populations despite pediatric trials being conducted.⁸² Among those 32 drugs, more than half (56%) failed to demonstrate effectiveness, 22% did not demonstrate sufficient safety, and 22% had both effectiveness and safety insufficiency. An inadequate study design, such as insufficient control of confounding variables, inappropriate dose and study endpoint selection, lack of a control arm, or small sample size, was the common reason for failed pediatric trials. A further evaluation of the failure rate and reasons associated with the failed pediatric trials in oncology between 2018 and 2025 is out of scope for this review.

Between 2018 and 2025 (Table 1), the average interval between adult and pediatric indication approval was 5.2 years. The longest delay was observed for rituximab, with adult and pediatric approvals occurring in 1997 and 2021, respectively. Notably, 15 drugs received adult and pediatric indications concurrently at the time of NDA or BLA approval, reflecting increasing integration of pediatric development into earlier stages of oncology programs.

Dose-Finding Studies

Dose-finding Phase 1 oncology studies, in both pediatric and adult populations, typically include a dose-escalation phase followed by a dose-expansion phase. The primary objective of dose-escalation is to systematically characterize safety, tolerability, and PK across small, escalating dose cohorts. The subsequent dose-expansion phase aims to refine the dosing schedule and further define toxicity and PK profiles, while providing preliminary evidence of antitumor activity to inform later-phase development.

Often in pediatric Phase 1 studies, dose escalation may be substantially streamlined when exposure–response relationships can be extrapolated from adult data and when disease biology and clinical course are expected to be similar between pediatric and adult populations (Figure 3). Under these circumstances, an initial cohort focused on PK assessment can be sufficient to identify a pediatric dose that achieves adult-comparable therapeutic exposure, followed by a single expansion cohort without the need to explore multiple dose levels. Furthermore, modeling and simulation to match PK in adolescent patients to adults in the absence of clinical data in adolescents of the same indication have led to approval for several drugs in this age group, including nivolumab-relatlimab FDC, vorasidenib, and cabozantinib due to difficulty of enrollment in pediatric populations.^{13,20,21} Such exposure-driven pediatric trial designs align with regulatory expectations under PREA, RACE, and PIP frameworks, particularly when pediatric development is initiated early and traditional sequential adult–then–pediatric development pathways are no longer feasible.

When disease similarity or exposure–response comparability between adults and children cannot be reasonably assumed, or when therapies are specific to pediatric populations, more comprehensive PK characterization and dose–finding programs are required prior to confirmatory safety and efficacy studies. Traditional rule-based 3+3 design is still commonly used for dose escalation (Table 3) due to its simplicity and conservative safety profiles, as it does not rely on the assumption of a dose–toxicity relationship. Because pediatric phase 1 oncology studies are frequently initiated after the adult recommended Phase 2 dose (RP2D) has been established, pediatric dose exploration is usually less extensive compared to adults, with the majority of two or three dose levels (Table 3).

To mitigate the prolonged escalation timelines and potential exposure to subtherapeutic doses inherent to traditional 3+3 design, alternative rule-based approaches such as accelerated titration¹²⁸ or rolling 6¹²⁹ designs have been increasingly adopted. Among 27 pediatric Phase 1 or Phase 1/2 dose-finding studies reviewed (Table 3), 13 employed a 3+3 design, 8 used

a rolling 6 design, 4 studies evaluated a single dose level (atezolizumab, daunorubicin and cytarabine liposome, granule formulation of selumetinib, and nilotinib), and 1 study (calaspargase pegol) tested two pre-specified dose levels. In most 3+3 studies, pediatric starting doses were set at one level below the adult RP2D or approximately 80% of the adult dose, whereas some studies, such as inotuzumab ozogamicin in Japanese pediatric, pembrolizumab, dordaviprone, and selpercatinib, initiated at the adult RP2D or approved dose using allometric scaling. A similar pattern was observed for rolling 6 designs, with most studies starting at adult-equivalent doses adjusted for body size (brentuximab vedotin in Japanese pediatric patients, blinatumomab, revumenib, nivolumab, larotrectinib, and selpercatinib), and two studies (inotuzumab ozogamicin and crizotinib) started at one level below adult dosing, while the other six started at adult RP2D or approved dose with allometric scaling. Ten drugs were approved without a dedicated pediatric dosing-finding study, and the majority of these approvals were limited to adolescents, with only one drug in children, three in infants, and one in neonates.

Bayesian designs like the continual reassessment method (CRM) or adaptive logistic design have been proposed. In recent years, model-based dose escalation designs, including modified toxicity probability interval (mTPI),¹³⁰ Bayesian Optimal Interval Design (BOIN),¹³¹ and new ruled-based designs such as interval 3+3 (i3+3)¹³² and backfill i3+3 (Bi3+3),¹³³ have been widely used in adult FIH trials. Compared with traditional 3+3 designs, these new designs have a higher likelihood of identifying true maximum tolerated dose (MTD) and assigning fewer participants to toxic dose levels. In the Phase 1 dose-finding and dose-confirmation study of pembrolizumab in pediatric patients (KEYNOTE-051), the mTPI study design was used to determine the pediatric RP2D (Table 3). This design allows flexible dose de-escalation from the initial dose of 2 mg/kg Q3W to a minimum of 1 mg/kg Q3W based on the proportion of patients with dose-limiting toxicities such that the target probability of toxicity is around 25% for the MTD.⁸⁶ With BOIN receiving FDA's designation for dose-finding trials,¹³⁴ we expect to see more implementation of these new designs in pediatric dose escalation trials.

Given the small sample size of the majority of pediatric oncology studies and the frequent inclusion of heterogeneous tumor types and age ranges, master protocols—particularly basket and umbrella trials—have gained popularity in recent years. Basket trials are particularly attractive for targeted therapies, as they evaluate treatments across multiple tumor types defined by shared molecular alterations or biomarkers rather than histology or anatomic location.¹³⁵ Selpercatinib, a

Table 3. Summary of Dose-Finding Study in Selected Oncology Drugs With US FDA Approval in Pediatrics Between 2018 and September 2025

Generic Name (Brand Name)	Study Name/ Trial Number (Phase of Study)	Study Objective	Study Age Group	Design (Sample Size)	Dose Level Tested	Starting Dose Justification	PK Sampling (Samples per Dosing Interval)
Approved in Adolescents and Older							
Ipilimumab (Yervoy)	NCT01445379 (Ph 1) ⁸³	Assess safety, PK, and immunogenicity	2-21 y/o	3+3 (N = 31)	1, 3, 5, 10 mg/kg Q3W	One level below the adult-approved dose	Intensive (N = 6)
Nivolumab (Opdivo)	ADVL1412 (NCT02304458) (Ph 1/2) ⁸⁴	Assess safety, tolerability, PK, MTD/TP2D, and efficacy of Nivo with or without ipilimumab	12 m/o-30 y/o	Rolling 6 (N = 13 in Ph 1; 63 in Ph 2)	3 mg/kg D1/15 Q28D (could be de-escalated to 1 mg/kg)	Equivalent to adult-approved dose (target C _{min} ≥ 10 ug/mL in 5/6 patients)	Intensive (N = 5)
Nivolumab and relatimab-rmbw (Opdivalag)	RELATIVITY-020 (NCT01968109) (Ph 1/2) ⁸⁵	Assess safety, tolerability, and efficacy	≥ 12 y/o	N.R. (varies in arms; no pediatrics were enrolled)	Rela SAV 20-800 mg Q2W; Nivo+ Rela SAV 80/20 mg Q2W to 480/1440 mg Q4W	NR	NR
Pembrolizumab (Keytruda)	KEYNOTE-051 (NCT02332668) (Ph 1/2) ⁸⁶	Assess RP2D, safety, and efficacy	6 m/o-17 y/o	Modified 3+3 (N = 12 in Ph 1; 143 in Ph 2)	2 mg/kg Q3W (could be de-escalated to 1 mg/kg or escalated to 10 mg/kg)	Equivalent to adult-approved dose (target toxicity probability of ~25% at MTD in the first 3 weeks and exposures <50% of adults)	Intensive (N = 4)
Pembrolizumab and berahyaluronidase alfa-pmph (Keytruda Qlex)	No dose-finding study in pediatrics was conducted. PK, safety, and tolerability of pembrolizumab SC was conducted in Ph 1, 3475A-C18 (NCT05017012) in adults with advanced solid tumors. ⁸⁷						
Lutetium Lu 177 dotatate (Lutathera)	No dose-finding study in pediatrics was conducted. NETTER-P (NCT04711135), a Ph 2 study, was to assess safety and dosimetry in adolescents (age 12-17 y/o) with adult dose (7.4 GBq/200 mCi Q8W). ⁸⁸						
Iobenguane I-131 (Azedra)	NCT00659984 (Ph 1) ⁸⁹	Assess MTD in neuroblastoma (no dose-finding study in PPGL)	1-30 y/o	3+3 (N = 16)	12, 15, 18 mCi/kg	One level lower than the previously identified DLT at 15 mCi/kg in neuroblastoma	NR (whole-body imaging was used for PK)
Belzutifan (Weireg)	No dose-finding study in pediatrics was conducted. LITESPARK-015 (NCT04924075), a Ph 2 study, was to assess efficacy and safety in age ≥ 12 y/o. LITESPARK-015 (NCT04924075), a Ph 2 study, was to assess efficacy and safety in age ≥ 12 y/o. Adult-approved dose (120 mg QD) was tested. ⁹⁰						
Tazemetostat (Tazverik)	No dose-finding study in pediatrics was conducted. Dose justified by RP2D from adult Ph 1 PK/PD and safety data. ⁹¹						
Vorarsidenib (Vorango)	No dose-finding study in pediatrics was conducted; approval was extended to ≥ 12 y/o via exposure matching (adolescents were eligible in the Ph 3 study, but only one aged 16 y/o enrolled).						
Cabozantinib (Cabometyx)	No dose-finding study in pediatrics was conducted; approval extended to ≥ 12 y/o via exposure matching.						

(Continued)

Table 3. (Continued)

Generic Name (Brand Name)	Study Name/ Trial Number (Phase of Study)	Study Objective	Study Age Group	Design (Sample Size)	Dose Level Tested	Starting Dose Justification	PK Sampling (Samples per Dosing Interval)
Approved in Children and Older							
Brentuximab vedotin (Adcetris)	NCT01780662 (Ph 1/2) ⁹²	Assess safety and efficacy	<30 y/o	3+3 (N = 46)	1.4, 1.8 mg/kg	One dose level below adult-approved dose	NR
	NCT01492088 (Ph 1/2) ⁹³	Assess safety, tolerability, RP2D, and efficacy	7-18 y/o	3+3 (N = 36)	1.4, 1.8 mg/kg	One dose level below adult-approved dose	NR
	JMA-IIA00229 (Ph 1) ⁹⁴	Assess safety and tolerability in Japanese pediatric patients	2-17 y/o	Rolling 6 (N = 6)	1.8 mg/kg	1.8 mg/kg was tolerated in pediatric patients in Western countries	NR
Atezolizumab (Tecentria)	GO29664/ IMATRIX (NCT02541604) (Ph 1/2) ^{95,96}	Assess safety, tolerability, PK, immunogenicity, and preliminary efficacy in pediatric and young adults with solid tumors	1-30 y/o	Single dose levels (N = 90)	15 mg/kg (max 1200 mg) for age < 18 y/o and 1200 mg for ages 18-29	Equivalent to adult-approved dose (1200 mg or 15 mg/kg for 80 kg); linear PK between 1-20 mg/kg	Sparse (N = 3)
Tagraxofusp-erz (Elzonris)	No dose-finding study in pediatrics was conducted. Safety evaluated in 1 child (2-12 y/o) and 2 adolescents (12-17 y/o). Efficacy was extrapolated from an adult study, STML-401-0114 (NCT02113982). Adult dose (12 µg/kg/day) was tested. ⁹⁷						
Selpercatinib (Retevmo)	LIBRETTO-001 (NCT03157128) (Ph 1/2) ^{98,99}	Assess MTD/RP2D (Ph 1), efficacy (Ph 2)	≥ 12 y/o	3+3 (N = 45)	20 mg QD, 20 to 240 mg BID	Adolescents were treated with adult RP2D (BW-based if < 50 kg)	Intensive (N = 5)
	LIBRETTO-121 (NCT03899792) (Ph 1/2) ^{100,101}	Assess safety/DLTs (Ph 1), efficacy (Ph 2)	2-20 y/o	Rolling 6 (N = 27)	92 mg/m ² QD	Equivalent exposure to adult RP2D and CYP3A4 matures at the age of 2	NR
Approved in infants and older							
Gentuzumab ozogamicin (Mylotarg)	NCT00053274 (Ph 1) ¹⁰²	Assess tolerability and efficacy	>0 y/o	3+3 (N = 29)	≥ 3y/o: 6, 7.5, 9 mg/m ² < 3y/o: 0.2, 0.3 mg/kg	NR (dose is 66.6% of adult dose for ≥ 3 y/o)	NR
Inotuzumab ozogamicin (Besponsa)	ITCC-059 (NTR5736) (Ph 1/2) ¹⁰³	Assess MTD/RP2D, safety, response, and PK/PD	1-18 y/o	Modified rolling 6 (N = 23)	1.4 mg/m ² with fractionated dose (de-escalated to 1.1 or 0.8 or escalated to 1.8 or 2.2)	Dose is 80% of the adult dose based on effective adult dosing, with projected similar PK in pediatrics	Intensive (N = 8)
	INO-Ped-ALL-1 (UMIN000033351) (Ph 1) ¹⁰⁴	Assess safety, tolerability, PK, immunogenicity, and efficacy in Japanese pediatric patients	1-17 y/o	3+3 (N = 6)	1.8 mg/m ² with a fractionated dose	Same as adult regimen	Intensive (N = 8)

(Continued)

Table 3. (Continued)

Generic Name (Brand Name)	Study Name/ Trial Number (Phase of Study)	Study Objective	Study Age Group	Design (Sample Size)	Dose Level Tested	Starting Dose Justification	PK Sampling (Samples per Dosing Interval)
Blinatumomab (Blinincyto)	MT103-205 (NCT01471782) (Ph 1/2) ¹⁰⁵	Assess safety, PK, RP2D, and preliminary efficacy	< 18 y/o (2-17 y/o in dose escalation, include <2 y/o before Ph 2)	Rolling 6 (N = 49)	5, 15, and 30 µg/m ² /day and a stepwise dosage of 15/30 µg/m ² /day (15 µg/m ² /day for the first 7 days and 30 µg/m ² /day thereafter)	Same as adult dose	Intensive (N = 4)
Naxitamab-ggqk (Danyelza)	Study 12-230 (NCT01757626) (Ph 1) ¹⁰⁶	Assess MTD	≥ 1 y/o	3+3 (N = 57)	0.3-3.2 mg/kg/day (0.9-9.6 mg/kg/cycle)	Based on prior clinical experience with the murine anti-GD2 antibody (m3F8) rather than toxicology data	Intensive (N = 8)
Rituximab (Rituxan)	No dose-finding study in pediatrics was conducted. PK in pediatrics aged ≥ 3 y/o (N = 35) was evaluated in NHL Study 11/ Inter-B-NHL Ritux 2010 (NCT01516580) with an adult regimen (375 mg/m ²). ¹⁰⁷						
Calaspargase pegol - mkn1 (Asparlas)	AALL07P4 (NCT00671034) (Bioequivalence) ¹⁰⁸⁻¹¹⁰	Assess PK comparability with pegaspargase	1-26 y/o	Two dose levels (N = 111)	2100 and 2500 IU/m ²	2100 IU/m ² was predicted to achieve comparable AUC _{0-25d} with pegaspargase	Intensive (N = 5)
Azacitidine (Vidaza)	No dose-finding study was conducted. No dose finding study in pediatrics was conducted. AZA-JMML-001 (NCT02447666), a Ph 2 study, was to assess PK, PD, safety, and activity. Dose was given at 75 mg/m ² . For BW < 10 kg or age < 1 y/o, the dose was 2.5 mg/kg to match adult exposure at the approved dose of 75 mg/m ² . Intensive PK (N = 7 or 5 for BW ≤ 20 kg was collected). ¹¹¹						
Daunorubicin and cytarabine liposome (Vyxeos)	CLTR0308-101 (NCT01943682) (Ph 1) ¹¹²	Assess safety and tolerability	1-21 y/o	Single dose level (N = 9)	Adult dose with option to escalate (cytarabine 134 mg/m ²) or de-escalate (cytarabine 67 mg/m ²)	Adult recommended dose (daunorubicin 44 mg/m ² and cytarabine 100 mg/m ²). BSA dosing achieves similar PK exposure in adults, and this dose level is tolerated in adults	Intensive (N = 10)
Dordaviprone (Modevso)	ONC014 (NCT03416530) (Ph 1) ¹¹³	Assess safety, PK, preliminary efficacy, RP2D	2-18 y/o	3+3 (N = 22)	125, 250, 375, and 500 mg QD based on weight group	Adult RP2D (625 mg) with allometric scaling to achieve similar PK exposure and good adult safety data	Intensive (N = 8)
Eflornithine (lwlfn)	NCT01059071 (Ph 1) ¹¹⁴	Assess safety, MTD, and PK	0-21 y/o	3+3 (N = 21)	500, 750, 1000, and 1500 mg/m ² BID	Prior adult chemoprevention trials showed acceptable safety and biological activity at 250-500 mg/m ² /day.	Intensive (N = 5)

(Continued)

Table 3. (Continued)

Generic Name (Brand Name)	Study Name/ Trial Number (Phase of Study)	Study Objective	Study Age Group	Design (Sample Size)	Dose Level Tested	Starting Dose Justification	PK Sampling (Samples per Dosing Interval)
Revumenib (Revufori)	SNDX-5613-0700 (AUGMENT-101) (NCT04065399) (Ph 1/2) ^{115,116}	Assess safety, tolerability, DLT, MTD/RP2D (Ph 1), and efficacy (Ph 2)	≥ 30 days	Rolling 6 (N = 68)	95 or 160 mg/m ² with and without a strong CYP3A4 inhibitor; respectively	PK matching between adults and pediatrics when using BSA dosing in <40 kg	Intensive (N = 7)
Treosulfan (Grafax)	No dose-finding study was conducted. No dose-finding study in pediatrics was conducted. MC-FludT.17/M (NCT02333058), a Ph 2 study, was to assess efficacy, safety, and PopPK to refine BSA-dependent dose recommendation. ¹¹⁷						
Selumetinib (Koselugo)	SPRINT (NCT01362803) (Ph 1) ¹¹⁸ SPRINKLE (NCT05309668) (Ph 1/2) ¹¹⁹	Assess MTD/RP2D, PK, safety Assess PK, safety, palatability, and efficacy of granule formulation	2-18 y/o	3+3 (N = 24) PK bridging (N = 36)	20, 25, and 30 mg/m ² BID Granules equivalent to 25 mg/m ² BID	50% of adult RP2D of 75 mg and preclinical toxicology Extrapolation from capsule formulation; expected equivalent exposure in young children per PopPK	Intensive (N = 10) Intensive (N = 9)
Crizotinib (Xalkori)	ADVL0912 (NCT00939770) (Ph 1/2) ¹²⁰⁻¹²²	Assess MTD/RP2D, safety, tolerability, PK (Ph 1), and efficacy (Ph 2)	1-21 y/o	Rolling 6 (N = 79)	100, 130, 165, 215, 280, and 365 mg/m ² BID	80% of adult MTD	Intensive (N = 8 or 9 on day 1; N = 5 at steady-state)
Dabrafenib (Tafinlar)	CTMT212×2101 (NCT02124772) (Ph 1/2) ¹²³	Assess RP2D for single and combination (Ph 1), efficacy (Ph 2)	1-17 y/o	3+3 (N = 48)	Dabrafenib: <12 y/o: 2.63 to 5.25 mg/kg/day ≥12 y/o: 2.25 to 4.5 mg/kg/day Trametinib: 0.0125, 0.025, 0.032, 0.04 mg/kg	Dabrafenib started at 50% of the previous single-agent pediatric RP2D, then escalated to 100%; Trametinib started at 50% of the adult RP2D and escalated via 3+3 with matching PK to adults	Intensive (N = 9)
Larotrectinib (Vitrakvi)	LOXO-TRK-15003 (NCT02637687) (Ph 1/2) ^{124,125}	Assess safety, DLTs, MTD/RP2D (Ph 1), and efficacy (Ph 2)	1 month-21 y/o	Modified rolling 6 (N = 32)	Dosing nomogram that matches 50, 75, 100, 150, 200, and 300 mg BID adult equivalent dose	Initially match adult PK exposure based on age and weight using PBPK, then BSA regardless of age	Intensive (N = 6)
Nilotinib (Tasigna)	CAMN107A2120 (NCT01077544) (Ph 1) ¹²⁶	Assess PK	2-17 y/o	Single dose level (N = 11)	230 mg/m ² BID	PK bridging to adult exposure; BSA-CL correlation supported BSA-based dosing	Intensive (N = 8)

(Continued)

Table 3. (Continued)

Generic Name (Brand Name)	Study Name/ Trial Number (Phase of Study)	Study Objective	Study Age Group	Design (Sample Size)	Dose Level Tested	Starting Dose Justification	PK Sampling (Samples per Dosing Interval)
Approved in Neonates and Older							
Emapalumab-Izsg (Gamifant)	No dose-finding study in pediatrics was conducted. NI-0501-03 (NCT010459562), a Ph I dosing finding study was conducted in adult healthy volunteers and NI-0501-04 (NCT01818492), a Ph 2/3 study was conducted in age < 18 y/o to evaluate efficacy and safety at 1 mg/kg IV Q3W, increased up to 10 mg/kg based on clinical condition. 1 mg/kg was predicted to inhibit > 99% of INF γ for 3 days based on PK, PD, and neutralizing concentration of emapalumab in adult healthy volunteers. ¹²⁷						

AE, adverse event; BID, twice daily; BSA, body surface area; CR, complete remission rate; DLT, dose-limiting toxicity; DoR, duration of response; EFS, event-free survival; MTD, maximum tolerated dose; NR, not reported; ORR, objective response rate; PBPK, physiologically based pharmacokinetic; PD, pharmacodynamic; PFS, progression-free survival; PK, pharmacokinetic; PopPK, population pharmacokinetic; PPGL, pheochromocytoma or paraganglioma; RP2D, recommended phase 2 dose; SAV, single agent vial.

selective RET inhibitor, was evaluated in LIBRETTO-001 basket study enrolling patients with *RET* fusion-positive or *RET*-mutant tumors across multiple histologies, supporting its tumor-agnostic approval (Tables 1 and 3).^{98,99,136} Similarly, larotrectinib, a tropomyosin receptor kinase (TRK) inhibitor, was developed through integrated adult (LOXO-TRK-14001 [NCT02122913]) and pediatric (LOXO-TRK-15003 [NCT02637687]) basket trials with seamless dose-escalation, expansion, and Phase 2 evaluation. Within this platform, body surface area (BSA)-based dosing and multiple formulations (capsule and liquid) were evaluated, and PK modeling enabled exposure matching across age from 1 month to 18 years.⁴⁴ By sharing infrastructure and enabling information flow across cohorts, such platform designs conserve scarce pediatric resources and speed up development, provided that tumor selection and combination strategies are carefully prespecified.

Model-Informed Dosing Strategies

Approval Experience

Dose selection in pediatric oncology is subject to heightened regulatory scrutiny because it frequently constitutes the primary evidentiary basis for extrapolation decisions and pediatric labeling under PREA, RACE, and the EU Paediatric Regulation. Pediatric dosing strategies must therefore balance between efficacy and safety while accounting for developmental changes in physiology across age groups.

Among the 33 drugs reviewed (Table 1), heterogeneity in pediatric dosing approaches was observed. Eleven drugs were approved for adolescents (≥ 12 years) with dosing aligned to adult regimens. Several adolescent approvals used fixed adult dosing, including nivolumab-relatlimab FDC (Opdualag), pembrolizumab subcutaneous formulation (Pembrolizumab Qlex), lutetium Lu-177 dotatate, and tazemetostat. Others apply body weight (BW) thresholds, most commonly 40 kg, to determine whether adolescents receive adult fixed dosing (≥ 40 kg) or BW-based dosing (< 40 kg). Pembrolizumab exemplifies a mixed strategy with BW-based dosing in adolescents (capped at the adult fixed dose) and fixed dosing in adults, whereas ipilimumab remains universally weight-based.

Four drugs approved for children aged ≥ 2 years retained adult-consistent BW-based dosing—brentuximab vedotin and atezolizumab (capped at adult doses), tagraxofusp-erzs (without a cap), and selpercatinib, which uses a 50-kg cutoff for adolescent tiered dosing and BSA-based dosing in younger children (with a minimum BSA threshold). Seventeen drugs were approved for infants and older populations (≥ 1 year for most; rituximab down to 6 months), of

which nine used BSA-based dosing. Several drugs employed mixed strategies contingent on age, BW, or BSA cutoffs, including gemtuzumab ozogamicin (0.6-m² BSA cutoff with multiple BSA tiers), blinatumomab (45-kg cutoff for fixed vs BSA-based), azacitidine (age- and BW-dependent transitions between BW and BSA dosing), revumenib (40-kg cutoff), and Larotrectinib (1.0-m² BSA cutoff). Additional agents—dordaviprone, cabozantinib, and dabrafenib—used weight-tiered regimens. Emapalumab, the only therapy approved from the neonatal period onward, employs BW-based dosing.

Population Pharmacokinetic Modeling

When disease biology and pharmacologic response are considered similar across age groups, exposure-matching is the primary rationale for pediatric dose selection (Figure 3). PopPK models incorporating allometric scaling are commonly used for adolescents and older pediatric patients to adjust for body size and composition.

Selpercatinib provides a representative example. PopPK modeling across about 500 participants demonstrated that adolescents weighing <50 kg would experience higher exposures at the adult dose (160 mg twice daily [BID]). A reduced dose of 120 mg BID achieved adult-comparable AUC and C_{max}, with BW identified as a primary covariate on PK; sex, race, and age did not meaningfully affect exposure. Exposure-QTc modeling further predicted that the higher exposures in lower-weight, for example, <50 kg adolescents could result in mean QTc prolongation exceeding 20 ms, supporting selection of the lower 120 mg BID dose. PK and safety data from adolescents (15-17 years) enrolled in LIBRETTO001 were consistent with those of adults, reinforcing this exposure-driven extrapolation approach in RET-driven thyroid cancer and MTC. Other use cases of PopPK to support pediatric dosing include calaspargase pegol-mknl and asparaginase *Erwinia chrysanthemi* in ALL.¹³⁷

An exposure-matching approach was utilized to support the approval of nivolumab-relatlimab FDC in adolescent patients with melanoma. Modeling and simulation predicted adolescent exposures relative to the adult efficacious targets, enabling efficacy extrapolation in pediatric subjects ≥12 years and ≥40 kg, where disease biology and expected exposure-response relationships for immune-oncology therapy are generally consistent with adults. In addition, safety data from Studies CA224047 and CA224020 showed comparable profiles across body weight (<50 vs ≥50 kg) and younger or older adults (aged 18-30 vs ≥30 years). These findings support exposure-matching and PopPK-based justification of adolescent dosing for nivolumab-relatlimab FDC when disease characteris-

tics and exposure-response relationships are similar to adults.

Therapeutic mAbs have demonstrated great success in numerous adult cancers. Although historically pediatric mAb PK differences were attributed primarily to body size, recent analyses indicate that tumor type and age may also influence clearance.¹³⁸ A PopPK analysis of nivolumab and ipilimumab demonstrated that, after adjusting for body size, pediatric patients aged 1-17 years with solid tumors or central nervous system tumors had lower clearance than adults, while clearance in pediatric lymphoma patients was similar. The identified tumor type-specific, age-related PK differences in pediatric patients are important considerations for optimizing dosing of mAbs and biologics in children.

Mechanistic Modeling Approaches

Mechanistic modeling approaches, such as PBPK and quantitative systems pharmacology (QSP), are particularly valuable for dose selection in younger children and infants, in whom rapid developmental changes in organ function, enzyme maturation, and body composition substantially affect drug ADME.

By incorporating age-dependent maturation of enzymes and transporters, disease-specific factors, and developmental pharmacology, PBPK models support rational dose selection in the youngest pediatric populations.¹³⁹ Barrett et al similarly emphasized PBPK as a mechanistic, bottom-up approach that overcomes the limitations of empirical dose adjustments and enables extrapolation to neonates and infants.¹⁴⁰ The authors outlined a workflow for pediatric PBPK applications, including dose selection and trial design evaluation, while noting varied confidence levels in PBPK modeling.

Subsequent evaluations further underscore PBPK's utility and areas for refinement.^{27,28} Zhou et al. demonstrated that PBPK models accurately predicted pediatric drug concentration-time profiles for most drugs studied, with 58 of 67 predictions within twofold and 43 within 1.5-fold of observed values—particularly for drugs primarily metabolized by CYP enzymes in children aged 1 month and older.¹⁴¹ In contrast, Leong et al, in reviewing PBPK models submitted to the FDA (midazolam, theophylline, and omeprazole), found that they generally underestimated clearance in neonates, infants, and young children,¹⁴² highlighting the need for improved model specification and verification frameworks.

PBPK modeling has been increasingly accepted by regulators for the prediction of pediatric DDI when clinical studies are not feasible. DDI risk in children differs from adults due to the ontogeny of CYP enzymes and transporters. By integrating in vitro and clinical data in adults with age-dependent physiology

(e.g., enzyme maturation, organ size, plasma proteins, and renal function), PBPK informs dose adjustments in pediatric patients and supports regulatory submissions, as illustrated by dordaviprone, for which PBPK was used to predict interactions with CYP3A modulators and transporter substrates.^{37,143}

Integrated and Emerging Modeling Approaches

Modeling approaches are complementary rather than exclusive. Integration of PBPK, popPK, and optimal design strategies has been shown to improve dose selection and study efficiency in pediatric oncology.¹⁴⁴ For example, combined modeling approaches accurately predicted pediatric docetaxel exposure and informed PK bridging strategies. More recently, revumenib applied an integrated MIDD framework: PopPK showed exposure similarity between adults and pediatric patients using BSA-adjusted doses; PBPK confirmed exposure alignment across children aged 1-17 years and with adults; and exposure–response analyses revealed plateaued efficacy with increased safety risks at higher exposures, underscoring the importance of precise pediatric dose selection.³⁹

Regulatory guidance also supports Bayesian approaches to borrow adult information when extrapolation is justified. Although broader application in pediatric oncology is still emerging, proof-of-concept examples across therapeutic areas illustrate how Bayesian and QSP frameworks can enhance the robustness of pediatric bridging strategy and dose selection, with continued refinement needed to support routine decision-making.^{134,145}

When Adult Exposure Matching Is Not Appropriate for Pediatric Dosing

Although exposure-matching has been a cornerstone to select pediatric dose, particularly for small molecules, there are scenarios where extrapolation may fail or be insufficient. For example, for drugs, largely mAbs, exhibiting target-mediated drug disposition (TMDD), integration of PK with PD and target-occupancy considerations are often necessary to optimize pediatric dosing because changes in PK parameters are less than body weight-proportional, potentially stemming from different target expressions between adults and children.¹⁴⁶ Simulations demonstrated that different dosing strategies may be needed depending on how to saturate the target expression—BW-based dosing is preferable for achieving consistent target concentrations, while fixed dosing may be more appropriate when targeting the same absolute amount.¹⁴⁶

Another example where extrapolation may not be sufficient is the development of chimeric antigen receptor T-cell (CAR-T) therapy in children. Unlike small molecules or biologics, CAR-T dosing in pediatric

patients is not based on achieving exposure comparable to that of adults. Although dosing is generally scaled by body weight, the effective exposure is determined by in vivo proliferation and persistence of the cells, which are influenced by disease burden, lymphodepletion, and CAR design rather than the administered dose itself.¹⁴⁷

Furthermore, when a drug has a narrow therapeutic range with a large interindividual PK variability, simply relying on adult exposure-matching may be insufficient and requires an individualized dosing approach. Alectuzumab is an anti-CD52 mAb that is widely used as part of conditioning regimens to induce profound immunosuppression. The pediatric study illustrates that standard weight-based dosing in pediatric patients does not result in consistent systemic exposure, with substantial interindividual variability driven by factors such as body weight and baseline lymphocyte count.¹⁴⁸ Moreover, clinical outcomes were strongly associated with exposure rather than administered dose, indicating that achieving a predefined exposure—such as one derived from adults—may not ensure optimal benefit–risk in pediatric populations. Instead, the authors developed a PopPK model for individualized precision dosing in pediatric allogeneic HSCT for nonmalignant diseases.¹⁴⁸

Other scenarios where adult exposure-matching may fail or be insufficient include when children have cancer types that differ from those typically seen in adults, when pediatric tumor biology differs from adult disease, when the pediatric indication involves ongoing, simultaneous treatments, when evaluating anti-cancer agents by adding the treatment to existing regimens, or when long-term safety issues are not captured by short-term PK equivalence. In these situations, additional efficacy and safety studies in pediatric populations will be needed to elucidate PK and response to treatment (Figure 3). Mechanistic modeling approaches with consideration of a broader set of criteria, including factors unique to pediatric physiology, disease characteristics, and potential interactions with other chronic therapies, could be considered to aid starting dose selection in pediatric populations.

Biopharmaceutical and Formulation Challenges

As outlined in earlier sections on regulatory expectations and the role of clinical pharmacology in pediatric oncology drug development, biopharmaceutical and formulation considerations are equally critical for enabling safe and effective pediatric dosing. Age-appropriate formulations frequently become binding elements within agreed pediatric development plans, influencing development timelines, post-marketing commitments, and ultimately pediatric labeling. Both the FDA PREA guidance⁶⁶ and ICH E11(R1)¹⁴⁹

Table 4. Oral Formulation Challenges in Pediatric Oncology

Formulation Challenge in Pediatric Oncology	Key Issues in Pediatric Cancer Patients	Case Examples
Age-appropriate formulations	Young children often cannot swallow tablets/capsules; they need liquids, minitabs, or dispersible forms; oncology drugs often have poor taste or stability.	Revumenib—Multiple formulations studied (capsule, tablet, and oral solution) in Ph1/2 studies. FDA approved tablet formulation only and requested bridging data of an oral solution as a postmarketing commitment because pediatrics or some adults may not be able to swallow the whole tablet. ^{39,151} Crizotinib oral solution was not palatable in children with refractory solid tumors and anaplastic large cell lymphoma. Crizotinib microspheres were palatable but intolerable due to toxicity. ¹⁵² Capsules and oral pellets are the approved formulations. ⁴²
Bioavailability differences across formulations	Pediatric physiology (gastric pH or motility) can affect dissolution and exposure; bridging studies required.	Alisertib enteric-coated tablet (exclusively studied in ADVL0921) had 68% relative bioavailability compared to power-in-capsule (exclusively studied in ADVL0812) in pediatric patients due to pH-dependent solubility. ¹⁵³ Larotrectinib achieved comparable C_{max} and AUC between capsule and liquid formulation across pediatric age groups, enabling approval of both formulations. ⁴⁴
Excipient-related toxicity or altered absorption	Children with cancer may have compromised organ function (hepatic/renal), increasing vulnerability to excipient toxicity; excipients may alter drug absorption.	Temozolomide oral suspension extemporaneously compounded—flavoring and suspending agents require safety justification in young children. ¹⁵⁴
Food-effect challenges in oncology drugs	Feeding patterns disrupted during chemotherapy (mucositis or nausea); high-fat meals in adult FE studies are not relevant; may alter exposure.	Ibrutinib oral suspension (pediatric studies) required reassessment of FE relative to adult capsules. ¹⁵⁵
DDIs and enzyme ontogeny affecting oral oncology exposure	CYP3A maturation; drug interactions with antifungals (azoles) and anticonvulsants common in oncology; PBPK modeling recommended to avoid high-risk clinical studies.	Imatinib pediatric PBPK models used to predict age-dependent CYP3A effects and guide dosing. ¹⁵⁶
Difficulty administering oral agents during intensive therapy	Mucositis, vomiting, and feeding tubes necessitate alternative formulations; crushing tablets may alter exposure or safety.	6-mercaptopurine often administered as an oral liquid formulation in children receiving ALL maintenance therapy, although formulation stability, shelf-life, and palatability have posed challenges. ^{157,158}

ALL, acute lymphoblastic leukemia; FE, food-effect; IM, intramuscular; IV, intravenous; PBPK, physiologically based pharmacokinetic; SC, subcutaneous.

emphasize the need for appropriate pediatric formulations, scientifically justified extrapolation strategies, and careful assessment of excipients and dosing approaches in children. Common challenges and examples relevant to pediatric oncology oral formulation are summarized in Table 4. Young children, especially neonates and infants, exhibit distinct gastrointestinal (GI) physiology that affects drug dissolution, gastric pH, gastric emptying, activity of bile and pancreatic fluid, enzyme maturation, and intestinal permeability.¹⁵⁰ These developmental differences make age-appropriate formulations essential. FDA and ICH guidance highlight considerations such as swallowability, palatability, manipulation feasibility, and dose flexibility.

Despite clear regulatory expectations and financial incentives such as pediatric exclusivity, the development and sustained availability of age-appropriate formulations remain limited. Quantitative evidence shows that

fewer than half of pediatric drug development programs are supported by age-appropriate formulations, with only 42.4% of studies using an age-appropriate formulation and 56.9% relying exclusively on adult solid dosage forms, even in incentivized settings.¹⁵⁹ This reliance persists in younger populations where formulation needs are greatest, including exclusive use of adult formulations in 26.3% of studies in children under 2 years and 44.9% in those aged 2-5 years. Consistent with these findings, broader analyses indicate that only approximately half of drugs have an age-appropriate formulation available at the time of labeling, highlighting a persistent gap between regulatory intent, incentive structures, and real-world implementation.¹⁵⁹

Bridging strategies are often required when pediatric-specific formulations differ from adult formulations. For example, trametinib included a pediatric oral solution with relative bioavailability (rBA) evaluated in adults to bridge systemic exposure

between formulations. Stability is particularly important for liquid formulations, which may degrade more rapidly than solids. In addition, excipients tolerated in adults (e.g., benzyl alcohol, propylene glycol, sorbitol, and ethanol) may pose age-specific safety risks, necessitating reformulation or reduced concentrations.

From a clinical pharmacology perspective, rBA studies are typically required when formulations differ, accounting for age-appropriate physiology, dose proportionality, and administration practices (e.g., tablet dispersion and capsule opening). Children have different feeding patterns and meal compositions relative to adults, which can influence drug absorption. Regulatory guidance therefore recommends evaluating food effects when dietary patterns differ in ways that may alter exposure, particularly in children <6 years of age, where GI physiology and diet differ substantially from adults.¹⁶⁰ Standard adult high-fat meals are often inappropriate for children; age-appropriate meal compositions should inform pediatric assessments. Modeling approaches, including PBPK, may reduce the need for dedicated pediatric food–effect studies but remain limited by uncertainty in younger populations.¹⁶¹ The PBPK model developed for entrectinib, a kinase inhibitor approved in pediatric patients older than 1 month of age with *NTRK* gene fusion-positive solid tumors, illustrated the challenges.¹⁶² The PBPK incorporated a mechanistic oral absorption model and was able to capture the observed PK in pediatric age between 4- and 20-year-old by accounting for the solubility changes due to the GI bile salt.¹⁶³ However, the model overestimated the exposure in patients less than 4 years old, potentially because of the larger uncertainty in the absorption process and CYP3A ontogeny.¹⁶²

Although oral administration is generally preferred in pediatric populations due to convenience and its noninvasive nature, many oncology drugs such as mAbs, ADCs, and cytotoxic chemotherapies are given intravenously. In these cases, adult drug concentrations are typically appropriate for pediatric patients, and additional formulation development may not be required. However, given the frequent use of body-size-based dosing, careful consideration should be given to dose-appropriate packaging, including single-use vials optimized for pediatric dosing to minimize drug wastage and support accurate administration.

Novel Approaches and Data Types

The increasing regulatory acceptance of real-world data, external controls, and adaptive evidence generation reflects pragmatic responses by health authorities to the constraints imposed by mandatory pediatric development requirements in rare oncology populations.

The rarity, heterogeneity, and geographic dispersion of pediatric tumors demand approaches that maximize information generation per participant while minimizing burden. Consequently, an expanding methodological toolkit—including RWD, synthetic and external control arms, MIDD, analytic advances in PK, and emerging applications of artificial intelligence (AI)—is rapidly reshaping feasibility and evidentiary expectations in pediatric oncology.

Regulators have explicitly supported innovative trial designs, such as basket/umbrella/platform approaches and adaptive designs, to make rare-cancer programs feasible while preserving inference quality. When randomized trials are impractical, rigorously curated external comparators can contextualize outcomes from single-arm pediatric studies, particularly when methods such as propensity matching and prespecified sensitivity analyses are applied.¹⁶⁴

Recent pediatric approvals illustrate both the potential and limitations of these approaches. In the development of eflornithine for high-risk neuroblastoma, the multicenter NMTRC003b Phase 2 trial relied on a propensity-matched external comparator derived from the ANBL0032 dataset to provide primary efficacy evidence, demonstrating how external controls can inform regulatory decisions when randomization is impractical.³⁸ For larotrectinib, RWD comparisons in TRK-fusion cancers consistently showed improved survival outcomes relative to SOC, providing evidence that complemented its seamless, tissue-agnostic basket-trial framework.⁴⁴ More recently, the approval of caplacizumab-yhdp in pediatric patients 12 years of age and older with a rare, acquired thrombotic thrombocytopenic purpura was based on an observational, retrospective chart review study with PK/PD simulation demonstrating the caplacizumab-yhdp exposure and suppression of von Willebrand factor antigen in pediatric patients were similar to adults at the approved dose.¹⁶⁵ These examples demonstrate how the novel approaches can accelerate drug approval in pediatric patients when the disease is rare.

Regulators support strategies to accelerate pediatric evidence generation through parallel enrollment of adolescents in adult trials, once initial adult safety, PK, and exposure–response data are available. This strategy leverages biological and developmental continuity between adolescents and adults and is supported by regulators when justified by MIDD and disease similarity.⁷⁹ Parallel enrollment reduced redundant pediatric dose-escalation studies and accelerated access to investigational agents in high-need indications.

Advances in AI/machine learning (ML) tools offer additional opportunities to streamline pediatric oncology development, and the FDA has encouraged these innovations to promote drug development.¹⁶⁶

Several AI/ML applications have been successfully utilized to facilitate data analysis for NDA or BLA, for example, outcome prediction, covariate selection/confounding adjustment, anomaly detection, and RWD phenotyping.¹⁶⁷ In pediatric oncology, AI/ML has mainly been used to perform risk stratification, predict treatment response, and optimize dose.^{168,169} Although at its nascent stage, AI-enabled medical devices developed for pediatric populations are expected to increase, and these devices can improve disease diagnosis or serve as a clinical decision support tool to improve productivity and confidence of healthcare providers.¹⁷⁰ Other potential applications of AI/ML include enhanced site- and patient-finding for ultra-rare molecular subsets, automating eligibility screening using structured and unstructured clinical data, and improved tumor-response assessment through ML-assisted imaging. AI/ML can also support adaptive MIDD frameworks by refining exposure predictions and dose selection as data accumulate, although transparent validation and regulatory oversight remain essential.

Finally, close collaboration with regulators facilitates fit-for-purpose study designs (e.g., seamless or platform architectures) and shared control arms. Recent FDA guidance outlines acceptable approaches for decentralized clinical trials, including tele-consent, home-health visits, and remote procedures, which may reduce geographic barriers and improve equity in trial access. Complementary guidance on digital health technologies enables the use of validated remote measurements for activity, sleep, neurological function, or symptom burden that are particularly relevant to pediatric oncology patients.¹⁷¹ Together, these regulatory-enabled innovations support more efficient, patient-centered pediatric oncology drug development while preserving scientific rigor.

Conclusions and Future Directions

Future directions in pediatric oncology drug development increasingly focus on accelerating access to targeted therapies through molecular profiling, transitioning from histology-driven strategies to tissue-agnostic approaches, and leveraging advanced modeling to optimize dosing. Key priorities include the rapid advancement of immunotherapies, expansion of precision-medicine trials, and strengthened global collaboration to address rare and high-risk pediatric cancers. Drug development is progressively guided by shared oncogenic driver mutations across pediatric malignancies rather than by the organ of origin alone, enabling broader application of targeted therapies. These efforts extend beyond kinase inhibition to include emerging

targets such as transcription factors and epigenetic regulators.

Precision-medicine trials, including initiatives such as Pediatric MATCH,¹⁷² play a central role in matching targeted therapies to specific genomic alterations in rare tumors while minimizing the number of patients required for clinical evaluation. Continued innovation in cellular therapies, particularly CAR-T therapies, remains a major focus, alongside the growing potential of mRNA-based vaccines and gene therapies for the treatment of refractory childhood cancers. Given the rarity of pediatric cancers, future progress will depend heavily on sustained global partnerships among regulatory agencies, industry, and academic consortia, such as the Children's Oncology Group, to enable efficient, large-scale, and collaborative clinical studies.

Innovative clinical trial designs, including master protocols, model-based dose escalation, and the use of single-arm or externally controlled studies, have been successfully implemented to reduce development timelines, costs, and patient burden. The increasing adoption of MIDD approaches has further accelerated dose selection, dose justification, and regulatory decision-making in pediatric oncology.

Despite these advances, mortality rates for several pediatric cancers, including high-grade gliomas and certain sarcomas, remain unacceptably high. Moreover, persistent disparities in pediatric cancer incidence and outcomes across race, ethnicity, geography, and other social determinants of health continue to pose significant challenges. Continued research and the development of efficient, equitable therapeutic strategies remain critically needed to improve outcomes for all children with cancer.

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Author Contributions

Y.W., X.W., S.B., L.A., T.D., G.X., S.K., and X.Z. wrote the manuscript; Y.W., X.W., S.B., L.A., T.D., G.X., M.C., T.H., S.K., and X.Z. performed the research.

Conflicts of Interest

Y.W., S.B., X.W., T.D., L.A., G.X., S.K., and X.Z. are employees of Gilead Sciences, Inc. and hold stock in the company. M.C. and T.H. declared no competing interests in this work.

Disclosure

Microsoft 365 Copilot was utilized to facilitate the extraction and summarization of key characteristics of regulatory review documents.

Data Availability Statement

Data sharing is not applicable to this article as no new data were generated or analyzed. All data supporting the findings of this study were derived from publicly available literature and regulatory review documents.

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